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The Fourth Dimension in Nineteenth-Century Physics

By Alfred M. Bork *

. . . any real body must have extension in *four* directions: it must have Length, Breadth, Thickness, and Duration. . . there are really four dimensions, three which we call the three planes of Space, and a fourth, Time. . . it happens that our consciousness moves intermittently in one direction along the latter from the beginning to the end of our lives. . . this is what is meant by the Fourth Dimension though some people who talk about the Fourth Dimension do not know they mean it. It is only another way of looking at Time.

THE contemporary scientist might have some difficulty attempting to identify this quotation out of context. Not sounding like a treatise, the passage seems to come from one of the countless popularizations of the theory of relativity. It comes not *too* surprisingly from a work of fiction, for science fiction often uses the terminology of science. The quotation belongs in the first chapter of H. G. Wells' *The Time Machine*; the Time Traveller is explaining the basis for his projected attempt to "travel" in time. But still another detail springs a surprise for many. This novel was written in 1895, before Hermann Minkowski's papers on a space-time.

We tend to view the relativistic conception of four-dimensional space as a twentieth-century idea. Geometers are known to have been interested in multidimensional spaces, which occur in the work of Georg Riemann and Hermann Günther Grassmann, among others. Riemann's famous paper on geometry was written in 1854, but remained unpublished until 1857; the concept of multidimensional geometry occurs also in Hermann von Helmholtz's papers of about this time. Among the many mathematicians who were concerned with this development were Arthur Cayley and Christian Felix Klein. Sommerville¹ lists eighteen hundred papers on *n*-dimensional geometry in his bibliography of 1911, perhaps three-fourths of these written in the nineteenth century. But here we want to show that the physicist too was interested in these ideas. Most of the examples are isolated, and some taken individually are rather trivial. But considered as a group, they furnish evidence that the Minkowskian view of the universe had a prehistory in

* Reed College. The author wishes to thank Gerald Holton and I. Bernard Cohen for their advice and encouragement.

¹ D. M'L. Y. Sommerville, *Bibliography of*

Non-Euclidean Geometry including the Theory of Parallels, the Foundations of Geometry and Space of n Dimensions (London: Harrison and Sons, 1911).

late nineteenth-century physics. Even the work in mathematics is important here, for the contemporary separation between pure geometry and physical geometry was not distinct then; this “fairlyland of geometry” explored by the mathematicians was often considered as a possible universe. Even among physicists the concept occurred much earlier than the nineteenth century; both Jean le Rond d’Alembert (1754) and Joseph Louis Lagrange (1797) mention,² in mechanical contexts, the possibility of considering time as a fourth dimension.

But first let us return to H. G. Wells. In *The Time Machine*, Wells brings in an outside “authority” to support his arguments concerning time and the fourth dimension: “But some philosophical people have been asking why *three* dimensions particularly—why not another direction at right angles to the other three? and have even tried to construct a Four Dimension geometry. Professor Simon Newcomb was expanding this to the New York Mathematical Society only a month or so ago.” The casual reader may have felt that Wells was introducing this only for authenticity. But Newcomb was not a fictional creation, he was a prominent American astronomer. Although some prejudice may be shown for a native son, the headline of the *Halifax Herald* of 1897 gives some idea of Newcomb’s status: “On Top of the Ladder of Fame—Nova Scotia’s Gifted and Honored Son—Professor Simon Newcomb, the World’s Greatest Living Astronomer.”³ Even the speech is not Wells’ invention. In *Nature* of 1 February 1894, an article entitled “Modern Mathematical Thought”⁴ appears; it is the speech Newcomb delivered to the annual meeting of the New York Mathematical Society on 28 December 1893. Although the speech was also printed in the United States, *Nature* is the likely source for Wells. Newcomb calls his speculations on the fourth dimension, here as elsewhere, “the fairlyland of geometry”; his argument is similar to Charles Howard Hinton’s. “It is, therefore, a perfectly legitimate exercise of thought to imagine what would result if we should not stop at three dimensions in geometry, but construct one for space having four. . . . Add a fourth dimension to space, and there is room for an indefinite number of universes.” But Newcomb’s interest must be classified as mathematical rather than physical, as suggested by the word “fairlyland”:⁵

From this point of view of physical science, the question whether the actuality of a fourth dimension can be considered admissible is a very interesting one. All we can say is that, so far as observation goes, all legitimate conclusions seem to be against it. No induction of physical science is more universal or complete than that three conditions fix the position of a point. The phenomena of light shows that no vibrations go outside of three dimensional space, even in the luminiferous ether. If there is another universe, or a great number of other universes, outside of our own, we can only say that we have no evidence of their exerting any action upon our own.

² R. C. Archibald, “Time as a Fourth Dimension,” *Bulletin of the American Mathematical Society*, 1914, 20: 409–412.

³ *Halifax Herald*, October 9, 1897.

⁴ Simon Newcomb, “Modern Mathematical Thought,” *Nature*, 1894, 49: 325–329.

⁵ *Ibid.*

Neither here nor in his other discussions of the fourth dimension does Newcomb mention time as a dimension, so Wells employs him slightly out of context.

But Newcomb's speech did not furnish Wells with the four-dimensional terminology. One of his earliest compositions, the incomplete "Chronic Argonauts"⁶ used similar ideas in 1888, and in many ways is similar to *The Time Machine*. People also travel in time, and the physical argument is almost identical: "no form can exist in the material universe that has no extension in time—nothing stood between men and a geometry of four dimensions—length, breadth, thickness, and duration—but the inertia of opinion, the impulse from the Levantine philosophers of the bronze age." Wells says⁷ that "In the students' Debating Society—I heard about and laid hold of the idea of a four dimensional frame for a fresh apprehension of physical phenomena. . . ." Bergonzi⁸ traces his interest to a debate held on 14 January 1887: E. A. Hamilton-Gordon, a fellow student at the Royal College of Science, talked on the "Fourth Dimension," and explicitly considered time as a possibility. Soon after this, Hamilton-Gordon saw C. H. Hinton's "Scientific Romances" (to be discussed here later), and possibly Wells was also familiar with them. For our present purposes it is interesting to note Bergonzi's comment that "Discussions about the 'Fourth Dimension' seem to have been in the air in the eighties. . . ."

The Time Machine was reviewed⁹ in *Nature*, a journal not usually given to reporting on works of fiction; the unsigned evaluation comments favorably on his use of scientific data, "Ingeniously arguing that time may be regarded as the fourth dimension of which our faculties fail to give us any distinct impression. . . ." Wells continued to use the fourth dimension in his other work; he even was to use it in Newcomb's form of a plurality of universes. One very interesting thread running through these uses is a primary concern with evolution. The real interest in *The Time Machine* is not in the machine itself, which is only a device allowing Wells to explore possible futures of the human race, probably for both direct and allegorical reasons. Slightly earlier Wells had been a student of Thomas Henry Huxley. It is entirely possible that this combination of Darwin and space-time is not accidental. Once the theory of evolution provided man with a past and a future, "time" became much more important for man. We see this perhaps more clearly in Wells than in any other writer of his period.

MAXWELL

Now we turn to dimensionality in physical contexts. As already noted, some of these considered individually are trivial, but that they occurred at all, and in such widely differing situations, may indicate the general interest in the idea. James Clerk Maxwell was understandably one of the first

⁶ Bernard Bergonzi, *The Early H. G. Wells* (Manchester: Manchester University Press, 1961), Appendix, pp. 187–214. (New York: Macmillan, 1934), p. 172.

⁸ *Op. cit.*, pp. 31–34.

⁹ *Nature*, 1895, 52: 268.

⁷ H. G. Wells, *Experiment in Autobiography*

physicists mentioning the “fourth dimension,” as he was in many ways an advanced thinker compared with his contemporaries. His work in electromagnetic theory, now considered the most important development in the physics of the nineteenth century, received little attention and almost no further development for a period of some twenty years. In a letter ¹⁰ to C. J. Monro on 15 March 1871, he makes the following comments:

The peculiarity of our space is that of its three dimensions, none is before or after another. As is x , so is y , and so is z .

If you have 4 dimensions this becomes a puzzle, for — first, if three of them are in our space, then which three? Also, if we lived in space of m dimensions, but were only capable of thinking n of them, then 1st, Which n ? 2d, If so, things would happen requiring the rest to explain them, and so we should either be stultified or made wiser.

I am quite sure that the kind of continuity which has four dimensions all co-equal is not to be discovered by merely generalising Cartesian space equations. . . .

This letter is in reply to one ¹¹ from Monro, but in the part reproduced by Campbell and Garnett there is no apparent reference to dimensionality. That Maxwell says “things would happen” shows that he at once considers the physical aspects. It is tempting to read the last sentence quoted in the light of the fact that the Minkowski four-dimensional space is *not* a Cartesian (Euclidean) four-space, but this would be reading contemporary attitudes into Maxwell.

There are several references to dimensionality in Maxwell’s poetry also. The last of four stanzas of “To the Committee of the Cayley Portrait Fund” ¹² (1874) shows a possible source in Cayley for Maxwell’s interest:

March on, symbolic host! with step sublime,
Up to the flaming bounds of Space and Time!
There pause, until by Dickenson depicted,
In two dimensions, we the form may trace
Of him whose soul, too large for vulgar space,
In n dimensions flourished unrestricted.

Again, there is no real justification for assuming that the mention of “Space and Time” anticipates Minkowski. Another reference ¹³ is in the poem “To Hermann Stofkraft, Ph. D., The Hero of a recent work called ‘Paradoxical Philosophy.’” But again we see some of Maxwell’s mathematical parents here:

My soul is an entangled knot,
Upon a liquid vortex wrought
By Intellect, in the Unseen residing,
And thine doth like a convict sit,
With marlinspike untwisting it,

¹⁰ Lewis Campbell; William Garnett, *The Life of James Clerk Maxwell*, new edition, abridged and revised (London: Macmillan, 1884), p. 289.

¹¹ *Ibid.*, p. 287.

¹² *Ibid.*, p. 414.

¹³ *Ibid.*, p. 420.

Only to find its knottiness abiding;
 Since all the tools for its untying
 In four-dimensioned space are lying,
 Wherein thy fancy intersperses
 Long avenues of universes,
 While Klein and Clifford fill the void
 With one finite, unbounded homaloid,
 And think the Infinite is now at last destroyed.

Paradoxical Philosophy was the second of two anonymously published books by Balfour Stewart and P. G. Tait. The first, *The Unseen Universe*, had at the top of the title page a knot, the symbol of William Thomson's vortex atom.¹⁴ It is this "knot" that can only be untied in four dimensions. This result was common knowledge at the time; Tait says ¹⁵ in a book review that "Professor Klein of Munich, sometime ago showed, as is well known, that knots cannot exist in space of four dimensions."

"S"

In *Nature* ¹⁶ of 26 March 1885, a letter signed "S" appears concerning time as a fourth dimension. "Possibly the question, What is the fourth dimension? may admit of an indefinite number of answers. I prefer, therefore, in proposing to consider Time as a fourth dimension of our existence, to speak of it as a fourth dimension rather than the fourth dimension." To conceive of a four-dimensional entity one must find something like motion in three dimensions. "Such an idea is that of successive existence. . . . Let any man picture to himself the aggregate of his own bodily form from birth to the present time, and he will have a clear idea of sur-solid in time-space." The concept of a "time-line" occurs in a form like its post-Minkowski form. The last part is concerned with the properties of a cube in time-space. A concluding sentence—"These results agree with the statement in your article"—appears to refer to the review of C. H. Hinton's "Scientific Romances" ¹⁷ printed two weeks earlier. Archibald ¹⁸ pointed out that "S" appears to be unaware of the earlier ideas of d'Alembert and Lagrange. We cannot for certain identify "S," but a conjecture would relate him to the debating group H. G. Wells attended. The idea that there might be different ways of conceiving the fourth dimension also occurs in Hamilton-Gordon's speech.¹⁹ The example of the human body is used in *The Time Machine* as well as in the letter written by "S." But the Hamilton-Gordon speech occurred two years after the letter to *Nature*.

HEAVISIDE

The reference to dimensionality by Oliver Heaviside is easily the most

¹⁴ C. G. Knott, *Life and Scientific Work of Peter Guthrie Tait* (Cambridge: Cambridge University Press, 1911).

¹⁵ P. G. Tait, "Zollner's Scientific Papers," *Nature*, 1878, 17: 420-422.

¹⁶ "S," "Four-Dimensional Space," *Nature*,

1885, 31: 481.

¹⁷ "Scientific Romances," *Nature*, 1885, 31: 431.

¹⁸ Archibald, *op. cit.*

¹⁹ Bergonzi, *op. cit.*, pp. 31-34.

trivial of this collection. But again we find an important physicist aware of the concept. In one of his earliest papers²⁰ on electromagnetic theory, "Magnetic Force and Electric Current," originally published in 1882, he is considering the integral theorems (today called Green's Theorem, Stoke's Theorem, divergence theorem, etc.) which connect spaces of different dimensionality. After stating the results, he compares the theorems with respect to dimensionality. The last goes from two to three dimensions. "Here we stop. Space has but three dimensions, and we cannot make a fourth to please anybody. Starting with a scalar for points, we have vectors for lines and surfaces, and a scalar again for a volume. I am not able to fathom the nature of the corresponding magnitude to be integrated in four dimensions, but probably it does not matter." Although a negative view, it does show awareness.

VECTORS AND QUATERNIONS

Another physical situation where ideas of many dimensions are found is associated not with a single individual but with several. During the 1890's a debate took place, primarily in letters to *Nature*, concerning the relative merits of using vectors in a quaternion setting (Tait, Knott) and using vectors without quaternions (Heaviside, Gibbs); it was a question of their use in physics, not as mathematical entities.

In his first letter²¹ to *Nature* concerning the relative merits of vectors and quaternions, J. W. Gibbs suddenly brings in arguments of dimensionality:

How much more deeply rooted in the nature of things are the functions $S\alpha\beta$ and $V\alpha\beta$ [the scalar and vector products] than any which depend on the definition of a quaternion, will appear in a strong light if we try to extend our formulae to space of four or more dimensions. It will not be claimed that the notions of quaternions will apply to such a space, except indeed in such a limited and artificial manner as to rob them of their value as a system of geometrical algebra. But vectors exist in such a space, and there must be a vector analysis for such a space.

He mentions that vector addition and scalar product can be extended with no difficulty, and that only a "little change" is needed for the vector product.

Tait's reply²² shows that Tait cannot understand why Gibbs raises this question of dimensions. He regards the connection between quaternions and three-dimensional Euclidean space as an argument favoring the use of quaternions: "What have students of physics, as such, to do with spaces of more than three dimensions?" It is interesting to note Tait's consistency here; twenty years earlier he made a very similar remark:²³

²⁰ O. Heaviside, *Electrical Papers*, Vol. I (London: Macmillan, 1892), pp. 195-231.

²¹ J. W. Gibbs, "On the Role of Quaternions in the Algebra of Vectors," *Nature*, 1891, 43: 511-513.

²² P. G. Tait, "The Role of Quaternions in the Algebra of Vectors," *Nature*, 1891, 45: 608.

²³ P. G. Tait, "Mathematics and Physics," in *Report of the British Association, Edinburgh*, Notices and Abstracts, 1871: 1-8.

In the eyes of the pure mathematician, Quaternions have one grand and fatal defect. They cannot be applied to space of n dimensions, they are contented to deal with these poor three dimensions in which mere mortals are doomed to dwell, but which cannot bound the limitless aspirations of a Cayley or a Sylvester. From the physical point of view this, instead of a defect, is to be regarded as the greatest possible recommendation.

But in spite of this protest, Gibbs returns to the point in his next letter.²⁴ Here his purpose is to support the importance of Grassmann's work, since Tait had tended to consider it of less consequence than William Hamilton's development of quaternions. Grassmann's algebra works for spaces with an arbitrary number of dimensions, but the quaternion analysis does not: "Grassmann's algebra of points will always command the admiration of geometers and analysts, and furnishes an instrument of marvelous power to the former, and in its general form, as applicable to space of any number of dimensions, to the latter." And at the end of the letter: "As a contribution to analysis in general, I suppose that there is no question that Grassmann's system is of indefinitely greater extension having no limitation to any particular number of dimensions." Tait's reply does not mention dimensions.

One can understand Gibbs' interest in n -dimensional space through his work in statistical mechanics.²⁵ When Lynde P. Wheeler was preparing his biography²⁶ of Gibbs he discovered in the Yale library unpublished notes on "Multiple Algebra for Dynamics." Edwin B. Wilson²⁷ has studied these notes. The earliest date noted was 1893, the last 1899; hence the notes began shortly after the letters to *Nature*. The concern is with statistical mechanics. In the earlier sheets Gibbs introduces two n -dimensional vectors, one for the generalized coordinates, one for generalized momenta, using n unit vectors. In 1896 these are combined into a single $2n$ -dimensional vector—a point in phase space. Hence Gibbs' interest actually takes us beyond the intent of this paper, into n dimensions instead of four dimensions.

Several years later, after other letters, the fourth dimension again enters; now the antagonists are A. MacFarlane and C. G. Knott, and the interchange is similar. MacFarlane²⁸ says in part: "I invite his attention to pr. 93 of the 'Principles' where I have investigated the rules for the several partial products of any number of vectors in space of not more than four dimensions (and they may be easily extended to space of higher dimensions)." Knott²⁹ protests: "Again, I fail to see what 'physical considerations' have to do with mathematics of the fourth dimension." The "again" is curious here—Knott seems to identify himself with Tait. And MacFarlane³⁰ replies:

²⁴ J. W. Gibbs, "Quaternions and the 'Ausdehnungslehre,'" *Nature*, 1891, 44: 79–82.

²⁵ This interpretation was suggested by Arnold Arons.

²⁶ Lynde P. Wheeler, *Josiah Willard Gibbs, the History of a Great Mind* (New Haven: Yale University Press, 1951).

²⁷ Edwin B. Wilson, "The Last Unpublished Notes of J. Willard Gibbs," *Proceedings of*

the American Philosophical Society, 1961, 105: 545–558.

²⁸ A. MacFarlane, "Vectors vs. Quaternions," *Nature*, 1893, 48: 75–76.

²⁹ C. G. Knott, "Vectors and Quaternions," *Nature*, 1893, 48: 148–149.

³⁰ A. MacFarlane, "Vectors and Quaternions," *Nature*, 1893, 48: 540–541.

“Professor Knott states that he fails to see what physical considerations have to do with mathematics of the fourth dimension. It is evident, however, that his perceptions cannot be taken as a criterion of truth, for every type of product of four vectors is geometrically real excepting the one which supposes them all independent of one another.”

It is clear that the pure-vector school was more willing to consider a four-dimensional space than were their opponents. But it is not easy to understand why. For example, just what does MacFarlane mean by the last sentence quoted? Again, the concept was present in a context which did not demand it. Quaternions were considered in a four-dimensional guise but always in mathematical rather than physical discussions with only one minor yet interesting exception. The Japanese mathematician Schumkichi Kimura noted³¹ that Hamilton’s differential operator ∇ had only three components, the three derivatives with respect to the coordinates rather than the four one might expect of a full quaternion operator. As a fourth component he chose, for reasons not expressed, the derivative with respect to time, thus anticipating the “lor” operator Minkowski was later to introduce. But he did consider physical applications: “In studying McAulay’s differential linear vector function, I came across the nablas whose arguments are the sum of vectors, which led me to consider the same operator in its wider sense, thereby extending it to the case of quaternion argument with transformations that might be useful in attacking the problems of physics and geometry.”

GRAVITY AND THE ETHER

One major unsolved problem of nineteenth-century physics was gravitation. What had been wanted since Newton was a mechanical explanation of gravitation, a mechanism to produce an inverse square attraction. Particularly in England physicists saw something occult in the idea of “action at a distance,” so the vague notion of “something” intervening between the particles and transmitting the force was widely held, although perhaps never established respectably by today’s standards. The ether was this medium and came during the nineteenth century to acquire optical and electromagnetic aspects also. Therefore it is not too surprising to find that those who speculated on the uses of the fourth dimension considered its possibilities for explaining gravitation and the ether.

In the introduction to W. K. Clifford’s papers,³² H. J. Stephen Smith comments that “Clifford was above all and before all a geometer.” With his interest in Riemann it was not unnatural for him to apply the ideas of geometry to physics. We gather that these attempts were not successful; unfortunately, most of his speculation is contained in a two-page abstract³³ of a talk to the Cambridge Philosophical Society in 1870, “On the Space-Theory of Matter.” He ends by saying: “I am endeavoring in a general way to explain the laws of double refraction on this hypothesis, but have

³¹ S. Kimura, “On the Nabla of Quaternions,” *Annals of Mathematics*, 1896, 10: 127–155.

³² W. K. Clifford, *Mathematical Papers* (London: Macmillan, 1882), p. xxxvii.

³³ *Ibid.*, pp. 21–22.

not yet arrived at any results sufficiently decisive to be communicated." As there are no further known papers, one assumes that Clifford did not consider this pursuit rewarding. But the basic thesis of the talk was a very interesting speculation:

I hold in fact

(1) That small portions of space *are* in fact of a nature analogous to little hills on a surface which is on the average flat; namely, that the ordinary laws of geometry are not valid in them.

(2) That this property of being curved or distorted is continually being passed on from one portion of space to another after the matter of a wave.

(3) That this variation of the curvature of space is what really happens in that phenomenon which we call the *motion of matter*, whether ponderable or ethereal.

(4) That in the physical world nothing else takes place but this variation, subject (possibly) to the law of continuity.

The relevance to one aspect of contemporary physics is obvious; the talk has been mentioned several times by J. A. Wheeler and his colleagues in their papers on geometrodynamics.³⁴ It is certainly an astounding view to be found in the nineteenth century. Further, this was not the only concept of twentieth-century physics which Clifford anticipated. He also discussed discrete space-time,³⁵ a theme often suggested which has never gained general support. Three years later in a lecture³⁶ at the Royal Institution on "The Postulates of the Science of Space" he does not return directly to these ideas. But he does discuss the concept of curvature and considers the possibility of a universe with constant curvature and finite volume. There is a hint of his previous interest at the end: "In fact, I do not mind confessing that I personally have often found relief from the dreary infinities of homaloidal space in the consoling hope that, after all, this other may be the true state of things." Maxwell refers to this in one of the poems quoted above.

Karl Pearson tried to combine the ether and the fourth dimension to produce a theory of an atom somewhat related to Thomson's vortex atom.³⁷

In this theory an atom is conceived to be a point at which ether flows in all directions into space; such a point is termed an *ether squirt*. An ether-squirt in the ether is thus something like a tap turned on under water, except that the machinery of the tap is dispensed with in the case of the squirt. Two such squirts, if placed in ether, move relatively to each other, exactly like two gravitating particles, the mass of either corresponding to the mean rate at which ether is poured in at the squirt.

The modern reader is interested to note that this view also allows for "negative matter"—that is, for ether-sinks. "Now the reader may naturally ask: Where can we conceive the ether to come from when it pours in at

³⁴ C. W. Misner; J. A. Wheeler, "Classical Physics as Geometry," *Annals of Physics*, 1957, 2: 525-603.

don: Macmillan, 1886), pp. 76-84.

³⁶ *Ibid.*, pp. 260-280.

³⁷ K. Pearson, *The Grammar of Science* (London: Walter Scott, 1892), pp. 318 ff.

³⁵ W. K. Clifford, *Lectures and Essays* (Lon-

the squirt or prime-atom? . . . Out from our space through the ether-squirt, out through matter we in conception pass . . . to another dimensioned space." Pearson is aware of Clifford's idea also. But unlike Clifford he published material³⁸ developing his idea further and in greater mathematical detail. The squirt is considered to be a point, and the velocity flowing out is infinite there. Among the areas he attempts to explain with this view of the atom are chemical action, spectral analysis, optical phenomena, and elasticity.

The third Victorian using the fourth dimension in connection with the ether is W. W. Rouse Ball. In "A Hypothesis Relating to the Nature of the Ether and Gravity"³⁹ he begins, as many concerned with higher dimensions, by considering a two-dimensional analogy.

Now suppose that the bodies in our universe have a uniform thickness in a fourth dimension, and that in that direction our universe rests on a smooth homogeneous body or medium. The transmission of energy apparently without the presence of an intervening medium may be then explained by supposing that the energy is caused by vibrations transmitted by the supporting space. If the thickness in the fourth dimension of the supporting space is small and uniform the law of propagation would be that of the inverse square of the distance, as is the case: if the supporting space was of four dimensions the law would be that of the inverse cube.

If separate media are required for both gravity and electricity, fourth and fifth dimensions will be required. Ball interpreted spectroscopic information, as did most at the time, as showing atoms are constantly in vibration, this vibration being transmitted to the supporting medium. Mass is proportional to the intensity of the vibrations. In the various editions of *Mathematical Recreations and Essays*⁴⁰ Ball further discusses both his own and Pearson's schemes.

HINTON

Our last nineteenth-century writer interested in the fourth dimension is Charles Howard Hinton. Instead of the isolated references provided by our other scholars, he produced a considerable literature stretching over a period of twenty years. His concern is also with the physical aspects of the problem; Hinton asks repeatedly what would be the physical consequences if a fourth dimension existed? And he offers several differing answers, some simple and some speculative. From 1884 to 1888 he wrote a series of small pamphlets called "Scientific Romances." In the first,⁴¹ as in many of them, dimensionality is a theme. He generalizes from line segment to square to cube to the corresponding four-dimensional structure; it is unlikely that this was original, but Hinton appears to be addressing not the mathematician but

³⁸ K. Pearson, "Ether-squirts; Being an Attempt to Specialize the Form of Ether-Motion which forms an Atom in the Theory propounded in former Papers," *American Journal of Mathematics*, 1891, 13: 309-362.

³⁹ W. W. Rouse Ball, "A Hypothesis Relating to the Nature of the Ether and Gravity,"

Messenger of Mathematics, 1891, 21: 20-24.

⁴⁰ W. W. Rouse Ball, *Mathematical Recreations and Essays* (London or New York: Macmillan, 1892-1939).

⁴¹ C. H. Hinton, *Scientific Romances No. 1. What is the Fourth Dimension* (London: Swan Sonnenschein and Co., 1884).

the intelligent layman. The reasoning is mostly by analogy about what happens between two dimensions and three dimensions. Hinton is thinking of a Euclidean or Riemannian four-space, but he recognizes that time plays some role, inasmuch as one can learn about figures in four dimensions by studying the different three-dimensional intersections generated by their motion. "If the straight line were placed slantingwise in reference to the plane, and moved downwards, it would always cut the plane in a point, but that point of section would move on." He shows that the mechanical features of a world of "atoms" in three dimensions can be produced by moving lines in four dimensions; permanence, impenetrability, inertia, and conservation of energy hold.

This model is consistent but certainly not necessary. Yet he ends the pamphlet with an explicit physical question: "Is there anything in the world as we know it, which would indicate the possibility of there being an existence in four dimensions? No definite answer can be returned to this question. But it may be of some interest to point out that there are certain facts which might be read by the light of the fourth dimensional theory." The suggestion is surprising: "we suppose that there is a centre of attraction somewhere off in the fourth dimension, and that the gases, which we know are simply more mobile liquids expanding in every direction under its influence. . . . Then the density of a gas would be a measure of the relative thickness of it in the fourth dimension. . . ." ⁴² But he does not stress this possibility, and he is fully aware of its speculative nature.

The second Romance is concerned partially with the first and second laws of thermodynamics. But in the next ⁴³ his interest returns to dimensionality. He mentions E. A. Abbott's *Flatland*, originally published at about the same time, but characterizes his own approach as more interested in the "physical facts." There is nevertheless a similarity in the two plane worlds. Very practical problems are discussed—can one make a wheel and how does one attach a rope to a cart? The inverse square law is replaced by an inverse first power. In the following Romance ⁴⁴ mirror images are basic; the evidence for four dimensions is now the existence of right-handedness and left-handedness. Here electricity, to play a more important role later in Hinton's work, also enters. Somewhat later ⁴⁵ Hinton seems to have a concept similar to Clifford's, although he is not mentioned. "That which we call the aether is far more probably the surface of a liquid, and the phenomena we observe due to surface tensions." This theme of a supporting film in the fourth dimension returns occasionally.

Hinton's most ambitious attempt to use the fourth dimension in physics occurs just after the nineteenth century. ⁴⁶ Lord Kelvin's vortex atom sug-

⁴² *Ibid.*

⁴³ C. H. Hinton, *Scientific Romance No. III. A Plane World* (London: Swan Sonnenschein and Co., 1886).

⁴⁴ C. H. Hinton, *Scientific Romance No. IV. A Picture of our Universe* (London: Swan Sonnenschein and Co., 1886).

⁴⁵ C. H. Hinton, *A New Era of Thought*

(London: Swan Sonnenschein and Co., 1888), p. 66.

⁴⁶ C. H. Hinton, "The Recognition of the Fourth Dimension," *Bulletin of the Philosophical Society of Washington*, 1902, 14: 179–203; *The Fourth Dimension* (London: Swan Sonnenschein and Co., 1904), Ch. XIV.

gests the consideration of rotations in four-dimensional space. Rotation is about a fixed plane in four dimensions. One can have two such rotations at the same time—one which would leave fixed the plane of the first two coordinates, one the plane of the second two coordinates—and this is a peculiarity of four dimensions. If one combines two three-dimensional rotations about two different lines, the resultant is still a rotation around some third line, but the two rotations in four dimensions are independent. Two particular double rotations, composed of those already mentioned, are called *A* and *B*. “If four dimensions exist and we cannot perceive them because the extension of matter is so small in the fourth dimension that all movements are withheld from direct observation except those which are three dimensional, we should not observe these double rotations, but only the effects of them in three-dimensional movements of the type with which we are familiar.” So far Hinton’s direction is not clear, but the next section provides the clue: He proceeds to discuss a rotating cell model in four dimensions, strongly reminiscent of Maxwell’s second paper⁴⁷ on electromagnetic theory.

We are now in possession of some of the conceptions of four dimensional mechanics, and will turn aside from the line of their development to inquire if there is any evidence of their applicability to the processes of nature.

Is there any mode of motion in the region of the minute which, giving three-dimensional movements for its effect, still in itself escapes the grasp of our mechanical theories? I would point to electricity. Through the labors of Faraday and Maxwell we are convinced that the phenomena of electricity are of the nature of a stress and strain of a medium . . . there must be some mode of motion of a body or of the medium in virtue of which a body is said to be electrified.

If we fix on a mode of motion as a definition of electricity, we must have two varieties of it, one for positive and one for negative; and a body possessing the one kind must not become possessed of the other by any change in its position. . . . Let us trace out the consequences of defining positive electricity as an *A* motion and negative electricity as a *B* motion. The combination of positive and negative electricity produces a current. [A reminder that Hinton’s concepts are not ours.] . . . An *A* motion and a *B* motion produce rotation round a plane, which is in the ether a vortex round an axial surface. . . . the conception we must form of a closed current is of a vortex sheet having its edge along the circuit of the conducting wire. . . . This geometric image corresponds to the definition of an electric circuit.

So Hinton sought to explain the nature of positive and negative electricity and current by using the fourth dimension. One must remember that Maxwell’s contemporaries found the concept of charge most puzzling in his work on electromagnetic theory; even his supporters claimed not to understand it. Hence Hinton picked a major problem of his day, less artificial than his previous attempt at reducing a gas to a liquid, in view of the scientific climate of the period.

⁴⁷ J. C. Maxwell, “On Physical Lines of Force,” in *Collected Papers of James Clerk Maxwell*, Vol. 1 (Cambridge: Cambridge University Press, 1890).

COMMENTS

In reprise, the point of this paper is that the fourth dimension had been used in physical situations before Minkowski's famous considerations. Obviously there was little coherence; although some scientists were aware of others with similar interests, there are only isolated fragments and no continuing dialogue. Furthermore, most of the examples are minor and are outside the mainstream of the physics of the time. The concept of multidimensionality was inherited from the mathematicians, but it excited physicists.

These speculations also illuminate the processes occurring in physics in the late nineteenth century. They reflect something of the turmoil of Newtonian physics, although it is a different aspect from the positivistic operational questioning of Mach. R. P. Feynman comments that "One often hears it said that physicists at the latter part of the nineteenth century thought they knew all the significant physical laws and that all they had to do was to calculate more decimal places. Someone may have said that once, and others copied it. But a thorough reading of the literature of the time shows they were all worrying about something."⁴⁸ The level of speculation itself is interesting; it exceeds the respectable speculations allowed traditionally in mechanistic physics. Perhaps we find the greatest change in the concept of explanation itself. From 1687 to the late 1800's a satisfactory explanation was a mechanical explanation, usually involving a model defined within classical mechanics. Yet the concepts examined by our scientists—the atom, the ether, gravitation, gases, positive and negative electricity—are not explained by mechanical pictures. Even a weaker criterion is violated; an explanation is often characterized as the discussion of the unfamiliar using familiar terms, but an explanation introducing the fourth dimension certainly does not satisfy such a characterization. It seems not unfair to view the mode of speculation shown in this paper as the forerunner of the mode of speculation, based on abstract mathematical systems rather than on mechanical models, dominant in twentieth-century theoretical physics.

One is also tempted to see another transitional feature here, perhaps stretching the evidence a little far. Although present in no conscious sense, there is an overtone of a changing view of reality. The Newtonian believed in the immediate reality of sensory experience, as modified by the insights of Newtonian physics. Even the concept of the atom was dubious for many scientists. To ask if our universe is "really" four dimensional is to go beyond a view of reality allowable to classical physics. In the twentieth century such a view in its extreme form was to lead to questioning the ultimate usefulness of the concept of "reality" itself. This possibility—that science treats the "labyrinth with the empty center"⁴⁹—may well be the single aspect of philosophy of science most frightening to contemporary man.

⁴⁸ R. P. Feynman; R. B. Leighton; M. Sands, *The Feynman Lectures on Physics* (Reading, Mass.: Addison-Wesley, 1963), pp. 40–49.

⁴⁹ G. Holton, "Über die Hypothesen, welche

der Naturwissenschaft zu Grunde Liegen" in *Eranos-Jahrbuch 1962: Der Mensch, Führer und Geführter in Werk* (Zürich: Rhein Verlag, 1963).